Final Project Task: Hackathon

Mansoura Robotics | Advanced AI Track

Discussion Day: Saturday, October 24, 2026

Project Idea

Pick any idea you like. Build a complete Deep Learning / Computer Vision project, from defining the problem to the final result, using what we learned in the 6 Sessions.

What matters most is that you understand every line you wrote.

Rules

1. **No AI tools.** Do not use ChatGPT, Copilot, Claude, Gemini, or any similar tool for your code, report, or discussion preparation.
2. **If you use a CNN:** build the architecture yourself (Conv, Pooling, BatchNorm, Dropout, etc.). Do not import ready-made models like `ResNet`, `VGG`, or `EfficientNet`.
3. **If you use YOLO:** you must put real effort into fine-tuning:
   - Prepare and clean your own dataset (collect, label, or modify it).
   - Use thoughtful augmentation.
   - Try different hyperparameters (learning rate, epochs, image size) and compare the results.
   - Analyze metrics (mAP, Precision, Recall) and show examples of errors.
   - Running `yolo train` with default settings is **not accepted**.
4. **Allowed libraries:** PyTorch, NumPy, scikit-learn, OpenCV, Matplotlib, Pandas, and similar ones.
5. **Accepted ideas:** any Classification, Detection, Regression, or Sequence (RNN) project that solves a real problem. Robotics-related ideas are preferred.
6. **Any code you cannot explain means lost marks**, even if the project works.

AI Check (Important)

Attar and Mohy will review your work. **If they suspect you used AI, they will delete your notebook and you will have to rebuild it from scratch in front of them.**

- You will write the code live while they watch and ask questions.
- If you cannot rebuild it, you get **zero** in Code Understanding and Rules.

- The only way to be safe is to write it yourself and understand it.

What to Submit

- A clean GitHub repository (code + `requirements.txt`)
- A `README.md` with: the problem, the dataset, the architecture, the results, and how to run it
- A notebook or scripts for training and evaluation
- A saved model (`.pth` or `.pt`) that runs during the discussion

Discussion (about 30 minutes)

- **3 minutes:** Explain your idea in simple words. What problem does it solve, who needs it, and how could it become a product?
- **25-27 minutes:** Code walkthrough and technical questions.

There are no slides. Just talk.

Example questions:

- Why did you choose this activation function / optimizer / learning rate?
- What happens if you remove BatchNorm or Dropout?
- Explain backpropagation on this line.
- Why did your model overfit, and what did you do to fix it?
- Why did you choose this layer size or kernel size?
- What does this line in your training loop do?
- How would you turn this project into a real product?

Grading (100)

Part	Marks	How it is graded
Code & Theory Understanding	**40**	You explain your code line by line and answer questions from the Sessions
Technical Implementation	**15**	Architecture built from scratch (or real YOLO fine-tuning) and a clean pipeline
Training & Evaluation	**10**	Loss curves, correct metrics, train/val split, handling overfitting

Part	Marks	How it is graded
Business Idea (verbal)	**10**	Clear problem, target users, value, and how it could become a product
Code Quality & Documentation	**10**	Organized GitHub, clear README, clean code
Following the Rules (No AI)	**10**	Honest work, passing the live rebuild if asked
Experiments & Improvements	**5**	You compare different experiments and justify your choices
Total	**100**	

Penalties

- Using a pretrained model (e.g., a ready-made CNN): **lose up to 15 marks, and Technical Implementation can drop to zero**
- Not being able to explain a large part of your code: **zero in Code Understanding**
- Failing the live rebuild after an AI suspicion: **zero in Code Understanding and Rules**
- YOLO without real fine-tuning: **marks lost in Implementation and Experiments**
- Late submission: **lose 5 marks**

Final Grades

- **90-100:** Excellent, ready to represent the team in the Hackathon
- **75-89:** Very good
- **60-74:** Acceptable, needs improvement
- **Below 60:** Try again

Best Wishes

Ahmed El Attar